

Matrix inequalities and norms

The maximum permanent on a positive-semidefinite unitary orbit

Difficulty: extreme **Importance:** interesting to the community**Status:** Solved**Provenance:** explicit exact extremal problem; no conjectured optimizer is supplied**Historical rating rationale:** An exact formula for arbitrary spectra would resolve a longstanding permanent optimization barrier and inform prescribed-spectrum matrix optimization.

Resolution — exact algebraic determination, 12 September 2026

Solved by [Sidney Holden](#), Center for Computational Biology, Flatiron Institute, Simons Foundation. [Submission and verified affiliation](#). [Proof PDF](#) · [LaTeX source](#) · [Independent AI-agent audit: PASS](#).

Theorem 2.1 and Sections 3–5 give an exact finite algebraic prescription for every nonnegative real spectrum in every order. A specialized critical-value elimination polynomial gives finitely many candidates. Explicit Schur-character sums compute a spectral moment at a proved finite order, which selects the maximum even when some real critical values are not Hermitian-feasible. The prescription has no remaining matrix optimization or unevaluated limit, and meets the original exact-value criterion below. For arbitrary real inputs it uses exact ordered-field arithmetic and comparisons; the reference programs accept rational inputs.

Form of the answer: this is not a compact general structural formula, an efficient arbitrary-order solver, or a classification of all general-spectrum optimizers. Those stronger goals are not claimed. Theorem 8.1 separately gives a two-candidate structural formula for one exceptional eigenvalue; Sections 9–10 prove the transition threshold and every equality case. The earlier rank-one expansion and equal-support reduction from [Holden’s partial submission, PR #186](#), are retained and credited in the dossier. This submission adds the all-spectrum determination and sharper exceptional-spectrum results.

A separate Codex agent independently audited the complete argument and its correspondence with this unchanged target. This is informal AI-agent review, not external human peer review or formal verification. No Lean verification or historical-priority claim is made. Whether a structural formula is preferred for the historical problem is exposed for maintainer review; it is not silently added to this entry’s completion criterion.

Earlier partial contribution — 12 September 2026

Holden’s [earlier theorem](#) determined the maximum for every one-exceptional-eigenvalue spectrum by comparing explicit support-size polynomials. Its [independent review](#) and [submission record](#) remain available. That partial result preceded the all-spectrum determination above.

Problem statement

For every integer $n \geq 1$ and every real list $\lambda = (\lambda_1, \dots, \lambda_n)$ with $\lambda_i \geq 0$, determine the exact value of

$$M_n(\lambda) = \max_{U \in \mathcal{U}_n} \text{per}(U^* \text{diag}(\lambda_1, \dots, \lambda_n) U), \quad \mathcal{U}_n = \{U \in \mathbb{C}^{n \times n} : U^* U = I_n\},$$

where $\text{per}(C) = \sum_{\sigma \in S_n} \prod_{i=1}^n c_{i, \sigma(i)}$ and S_n is the permutation group. The permanent of these Hermitian positive-semidefinite matrices is real. Compactness ensures the maximum exists. A resolution should express its value from the prescribed eigenvalues for arbitrary order, rather than only bound the objective or restate its optimization definition.

Relevance

The eigenvalues fix an entire unitary similarity class while the permanent varies within it. Determining its extreme value is a concrete optimization problem over matrices with a prescribed spectrum.

References

1. F. Zhang, *An update on a few permanent conjectures*, *Special Matrices* 4 (2016), 305–316, passage headed “Marcus–Minc max-per- U problem 1965.” [Primary text](#); [journal](#).
2. J. H. Drew and C. R. Johnson, *Counterexample to a conjecture of Mehta regarding permanental maximization*, *Linear and Multilinear Algebra* 25(3) (1989), 253–254, the counterexample. [DOI](#).

Status check — 2026-09-10

Zhang’s latest arXiv record is v1 and explicitly lists the general maximization problem. Its statement and discussion of the equal-diagonal counterexample were inspected. The original 1989 full text was not retrieved; that counterexample’s role was checked through Zhang’s explicit account and bibliography. Searches used *maximum permanent unitary 2026*, *maximum permanent eigenvalues 2025 2026*, and *Marcus–Minc max-per- U problem*. No general spectral formula was located. The claim that an equal-diagonal representative always maximizes the permanent is false and is not imposed. The other catalog permanent entries concern inequalities between distinct matrix functions or products, not this exact orbit maximum.

Audit update (2026-09-10): Rechecked Zhang’s Marcus–Minc max-per- U statement and searched for later exact spectral formulas. Equal-diagonal optimality remains an invalid shortcut; no all-spectrum determination was located. This is a bounded literature check, not a proof that no solution exists.